Report Document

Modifications to Existing Footprint - Feasibility Study Report

Client Name:	Clare Crook	Date:	19th May 2026
Your Ref:	Modifications to Existing Footprint	Our Ref:	9802-RBP-ZZ-ZZ-RP-X-100000
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Document History

Revision No.	Date of Issue	Issued For	Revision Description	Author
P01	19 May 2026	For comment	First Issue	Quynh Hill

Contents

1.0 EXECUTIVE SUMMARY

This feasibility study confirms that the proposed Lyophiliser can be successfully integrated within Rooms PG.10, PG.11, and PG.12, subject to the inclusion of a new airlock arrangement. No constraints have been identified from a mechanical, electrical, HVAC, or civil perspective that would prevent delivery of a compliant design.

A key outstanding item relates to the overall length of the dehumidification unit, which will require further engagement with the vendor during the FEED stage to determine whether the footprint can be optimised.

The proposed HPLC installation in PG.05 is also considered feasible, with recent reductions in pump skid size improving operational access. No issues have been identified that cannot be resolved during the next phase of design development.

The primary programme driver is the Lyophiliser lead time, currently estimated at 8 to 10 months. Early placement of orders will be critical to maintaining the overall schedule, and further vendor engagement may present opportunities to reduce lead time.

Under the current sequencing strategy, and despite prioritising procurement of the Lyophiliser, a Q3 2027 operational start date for the HPLC remains achievable. In contrast, the Lyophiliser is more likely to reach operational readiness in Q4 2027.

2.0 INTRODUCTION

R B Plant Construction Limited (RBPC) have been engaged by Asymchem to undertake a feasibility study for the introduction of two new process systems at their Process Development Facility (PDF) in Discovery Park, Sandwich, UK:

- High Performance Liquid Chromatography (HPLC)
- Lyophiliser

This feasibility study has been conducted in accordance with discussions and decisions made in a series of project meetings, along with the requirements shown within the Building Layout drawing and equipment information documents supplied by Asymchem. Further clarifications on the requirements have been sought through Request for Information RFI001 and RFI / action log.

The purpose of this document is to summarise the findings of the feasibility study, with references to the design documents that contain the provisional key parameters of the proposed systems.

3.0 REFERENCE DOCUMENTS

This document should be read in conjunction with the following reference documents:

1) 9802-ZZ-ZZ-CP-X-100001 – EST Project Cost Estimate
2) 9802-RBP ZZ-ZZ-SH-V-300000 – Demolition/Relocation Equipment List
3) 9802-RBP ZZ-ZZ-SH-V-320001 – Equipment List
4) 9802- RBP-ZZ-ZZ-DR-V-340000 – HPLC Process Flow Diagram
5) 9802- RBP-ZZ-ZZ-DR-V-340001 – Process Flow Diagram
6) 9802-RBP-ZZ-ZZ-DR-E-500000 – MCC-P Single Line Diagram & Cable Schedule
7) 9802-RBP-ZZ-ZZ-SH-E-500001 – MCC-P New Circuit Schedule
8) 9802-RBP-ZZ-ZZ-M3-M-220000 – New Equipment Layout
9) 9802-RBP-ZZ-ZZ-RP-R-050000 – Risk Register

4.0 PROJECT MANAGEMENT

4.1 Project Cost Estimate

A Project Cost Estimate is presented as an MS Excel workbook (ref. 1). Costs at this stage are assessed to be within an accuracy range of ± 50%.

Feasibility stage engineering design work has been executed to allow the estimate to be generated on the following basis:

- Equipment and piping disinvestment costs estimated from a budget quote supplied by a potential contractor following a site visit;
- Electrical disinvestment costs estimated from an existing estimate database, with checks to ensure that unit costs reflect current conditions
- HPLC equipment cost as supplied by Asymchem China;
- Lyophiliser equipment cost as supplied by Asymchem China;
- Mobile head tank cost obtained via a budget quote from a potential vendor for a broadly similar tank specification adjusted for capacity and increased functionality;
- Waste tank cost obtained using a figure from similar equipment quoted on a previous project and updated to allow for inflation and differing duties;
- Pump costs obtained using figures from similar equipment quoted on previous projects and updated to allow for inflation and differing duties;
- Instrument costs obtained using figures from similar equipment quoted on previous projects and updated to allow for inflation and differing duties;
- Mechanical installation cost, including piping, valves, fittings, etc. obtained using an existing estimate database, with checks to ensure that unit costs reflect current conditions;
- Electrical installation cost obtained using an existing estimate database, with checks to ensure that unit costs reflect current conditions;
- Control system integrator costs estimated using a factor per I/O point based on previously executed projects of a comparable size/complexity, with adjustment for inflation;
- Civil / structural construction contractor materials and labour estimated from approximate bills of quantity, and applying current unit rates, experienced previously for this type of project in a similar field;
- Contractor's fees are in accord with the project execution strategy, presented below.

4.2 Project Execution Strategy

Further discussion will be required to develop the required project execution strategy. However, one possible sequence is as follows:

(i) Based on a comparison of the project cost estimate against the available budget, changes will be made to the feasibility study to give an agreed solution.

(ii) On approval of the feasibility study, Project Cost and Programme, Asymchem will agree upon a Procurement Strategy and submit a Project Capital Expenditure (CapEx) request to raise the required funds for the long lead items.
(iii) Front end engineering design (FEED) will be performed to give better definition of the project schedule, design of equipment items and the installation/disinvestment requirements. This should enable tendering of the main packages and early procurement of any long-lead items.
(iv) On approval of the FEED Study, Project Cost and Programme, Asymchem will agree upon a Procurement Strategy and submit a Project Capital Expenditure (CapEx) request to raise the required funds for the remaining equipment items works.
(v) Once CapEx approval is given then the project will proceed with a Tendering process in line with the Procurement Strategy
(vi) As a result of the tender process contracts will be awarded to progress with the EPCM (Engineering, Procurement, and Construction Management) phase, including the removal of any redundant equipment, reclaiming any items suitable for re-use. The relevant contractors will be appointed to finalise the design, assist in the procurement of equipment and award of construction contracts, and provide overall project management and construction support services. Some roles may be performed by Asymchem.
(vii) Equipment will be purchased by the nominated responsible contractor, or directly by Asymchem, depending on the Procurement strategy agreed.
(viii) Construction contracts will be awarded by Asymchem for:
- Mechanical installation works – installation of equipment items, valves, piping, supports, instruments, etc.
- Electrical & Instrumentation – supply and installation of new electrical hardware and instruments; supply and installation of distribution boards; installation of control panels; installation of new electrical, control/instrumentation and earth cabling and containment
- Control system integration – detailed design of control system; supply and configuration of control system hardware and panels; testing and commissioning.
- Civil/structural installation works

4.3 Project Programme

A high-level project programme has been developed to outline the critical path (see below). Preliminary lead times of 8 to 10 months have been indicated for the Lyophiliser, and 18 weeks for the HPLC, subject to completion of Design Qualification (DQ) and agreement of commercial terms.

The scope of works associated with the disinvestment and construction of new areas in PG.12 has been evaluated and is currently estimated to require 20 weeks for completion. Further discussion with the vendor is required; however, it is considered likely that this duration could be reduced to approximately 14 weeks.

The current proposal, subject to further development, is to prioritise the procurement of the Lyophiliser and HPLC to allow manufacturing to commence at the earliest opportunity. Early vendor engagement will also facilitate the release of detailed design information, enabling the engineering design to progress in parallel.

To enable demolition works to proceed, it is proposed to temporarily relocate the filter drier to PG.10. This approach minimises disruption to operations by reducing the outage duration of the filter drier. The programme should be sequenced such that demolition, airlock installation, and area segregation within PG.12 are completed in advance of returning the filter drier to its original location. This should occur prior to delivery of the HPLC to site, allowing installation in PG.10 to proceed with minimal impact on ongoing operations. The construction works can be segregated by the newly formed wall within PG.12, enabling continued operation of the filter drier during subsequent phases of the works.

Following delivery of the Lyophiliser, associated equipment for PG.12 (including the dehumidifier) will be brought in via the double doors local to the loading bay, with provision for panel removal if required. Installation will then commence at the earliest opportunity.

The current programme indicates that the HPLC will be available for use by August 2027, meeting the Q3 beneficial use target.

The Lyophiliser is scheduled to be operational in November 2027; however, any acceleration of this date will depend on achieving a reduced manufacturing lead time.

Note: This programme assumes that funding is available to enable prompt placement of vendor orders.

From a FEED and detailed design perspective, these activities are expected to align with the project critical path. The FEED stage is presently estimated at 12–14 weeks, followed by a detailed design phase of 18-20 weeks, subject to the timely provision of information from the main vendors.

4.4 High Level Project Programme

Long Lead Items	**26/05/26**	**380 days**	**08/11/27**
Procure Lyophiliser - Asymchem	**26/05/26**	**380 days**	**08/11/27**
Finalise Lyophiliser Specification	26/05/26	4 wks	22/06/26
Funding approved, commercial agreements and deposit paid	23/06/26	4 wks	20/07/26
Lyophiliser build	28/07/26	9 mons	05/04/27
FAT	06/04/27	5 days	12/04/27
FAT remediation actions	13/04/27	4 wks	10/05/27
Shipping (assume via boat)	11/05/27	6 wks	21/06/27
Installation	01/06/27	13 wks	30/08/27
Validation activities	**31/08/27**	**50 days**	**08/11/27**
Commissioning and SAT	31/08/27	4 wks	27/09/27
IQ / OQ	28/09/27	2 wks	11/10/27
PQ	12/10/27	4 wks	08/11/27
PQ complete	08/11/27	0 days	08/11/27
Procure HPLC - Asymchem	**18/08/26**	**255 days**	**09/08/27**
Finalise HPLC Specification	18/08/26	4 wks	14/09/26
Funding approved, commercial agreements and deposit paid	15/09/26	4 wks	12/10/26
HPLC manufacture	20/10/26	18 wks	22/02/27
FAT	23/02/27	5 days	01/03/27
FAT remediation actions	02/03/27	3 wks	22/03/27
Shipping (via air freight)	30/03/27	2 wks	12/04/27
Installation works in PG.05	20/04/27	6 wks	31/05/27
Validation activities	**01/06/27**	**50 days**	**09/08/27**
Commissioning and SAT	01/06/27	3 wks	21/06/27
IQ / OQ	22/06/27	3 wks	12/07/27
PQ complete	13/07/27	4 wks	09/08/27
Procure Mobile Tanks -RB Plant	**21/07/26**	**205 days**	**03/05/27**
Procure Waste Tank -RB Plant	**21/07/26**	**195 days**	**19/04/27**
Demolition	**24/12/26**	**81 days**	**15/04/27**
Relocate Filter dryer to PG.05	24/12/26	3 days	29/12/26
Demolition, formation of air lock, separation of PG.12, relocate vacuum pump	05/01/27	14 wks	13/04/27
Move filter dryer back to PG.12	13/04/27	3 days	15/04/27

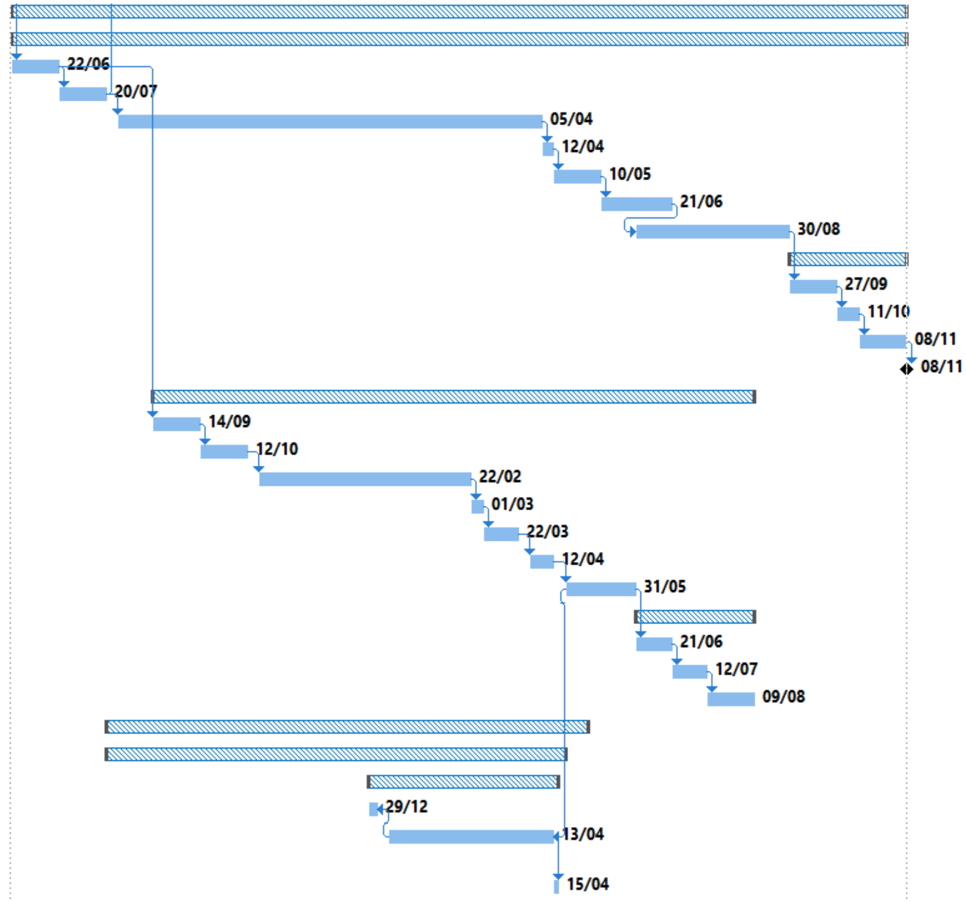

Figure 1 – Showing the Lyophiliser dictating the critical path

4.5 Project Risk Register

The items detailed below represent the highest-rated risks following the implementation of mitigation measures.

For Risk Item 11, Asymchem will progress the funding strategy internally.

For Risk Item 16, mitigation can be achieved by relocating the mobile spray dryer to PG.05, thereby avoiding disruption to ongoing operations. The programme sequencing can be further refined during the FEED stage.

Item #	Category Cost Time Scope Quality Safety Other	Sub-category	Primary Action by	Secondary Action by	Risk Rating (Unmitigated)			Status Initiated Reviewed Actions agreed Progressing Completed	Item Description *There is a risk that……*	Risk Rating (Mitigated)			Discussion, Mitigating Actions required/proposed
					Probability (unmitigated) from 1 to 5	Impact from 1 to 5	Rating			Probability (mitigated) from 1 to 5	Impact from 1 to 5	Rating	
11	Cost	Strategy			3	4	12		The funding strategy is not clear for the project	3	4	12	Client has a clear funding strategy Asymchem to score
16	Cost				5	5	25		Operation of filter driers are required during demolition work and installation of Lyophilizer	4	5	20	Not able to move spray drier to alternative position as new HPLC installation in that area. Currently can schedule spray dryer operation outside of demolition work. During build phase, this risk needs to be managed as there is currently only short term visibility of operational demands. Need to understand build timelines to fully assess probability

5.0 EQUIPMENT ITEMS

Equipment items for disinvestment or relocation are shown in Demolition Relocation Equipment List 9802-RBP-ZZ-ZZ-SH-V-320000.

New equipment items required are shown in Equipment List 9802-RBP-ZZ-ZZ-SH-V-320001. Further details of the equipment are given in the sections below.

Items not being supplied by Asymchem will be further defined during the FEED stage and described in Equipment Datasheets.

5.1 HPLC

HPLC will consist of three main items:

- DAC300 Column
- Slurry tank
- CP300 pump skid

All items will be specified and supplied by Asymchem.

5.2 Lyophiliser

The Lyophiliser will consist of the following main items:

- Freeze dryer chamber
- Freeze dryer condenser
- Refrigeration unit
- Isolator
- CIP Tank 1
- CIP Tank 2
- Dehumidifier

All items will be specified and supplied by Asymchem.

5.3 Mobile Head Tanks

Mobile head tanks are to be used for:

- feed material to HPLC via the pump skid (2 no.)
- fraction collection from HPLC column (3 no.)

Full details of the required mobile head tanks are to be defined.

Current suggested working capacities for the 5 no. mobile head tanks are 2 no. at 500 litres (feed) and 3 no. at 200 litres (fraction collection).

All mobile head tanks will be made from 316L stainless steel.

5.4 Waste Tank

Waste tank will be used for temporary holding of the front and tail fractions from the HPLC system. Material is held before being sent to the approved disposal route, but can be returned to the building for re-processing.

Full details for the waste tanks are to be defined.

Current suggested working capacity is 2,000 litres.

Material of construction to be 316L stainless steel.

5.5 Pumps

The following pumps are currently included in the design

- HPLC transfer pump
- HPLC waste pump
- Lyophiliser CIP pump
- Lyophiliser CIP waste pump

Details of the pumps are to be defined, and the duties are to be confirmed. All pumps are assumed to be in stainless steel.

5.6 Vacuum pump DVP01

The existing vacuum pump DVP01 in GP1 Ground Floor West Service Area, PG.12, is to be relocated. Cost has been allowed for this in the Mechanical Works.

6.0 MECHANICAL WORKS

Mechanical works will be further defined in the FEED stage and detailed in a mechanical installation specification.

6.1 Piping

For the purposes of the feasibility study, a preliminary assessment has been undertaken to identify pipework and services requiring disinvestment, and a budgetary quotation has been obtained from a vendor to support these works.

Estimated pipework lengths for new installations have been developed to inform the associated cost estimate.

During the FEED stage a disinvestment line list will be produced and identified at site including the drain down / isolation points.

For the new pipework, a line list will be developed during the FEED stage, supported by the creation of a 3D model to define the proposed routing. This will enable the generation of a material take-off, which will detail pipe lengths and fittings allowing for more accurate cost estimation.

6.2 Equipment Installation

Mechanical installation activities have been incorporated into the mechanical installation estimate, covering both free-issue equipment (including the Lyophiliser and HPLC) and all new equipment items identified within the equipment schedule. The scope also includes relocation of an existing vacuum pump. For the feasibility stage, a budget quotation was obtained from a specialist vendor to inform the cost estimate.

6.3 Instrument Installation

Mechanical installation of instruments has been included in the mechanical installation estimate.

6.4 Equipment Layout

The layout has been developed using information provided by the main equipment vendors. Based on this, a suitable arrangement has been defined to ensure safe and effective access for the operation and maintenance of the new equipment, while also maintaining the operability and maintainability of the existing equipment.

6.4.1 Lyophiliser Equipment Layout

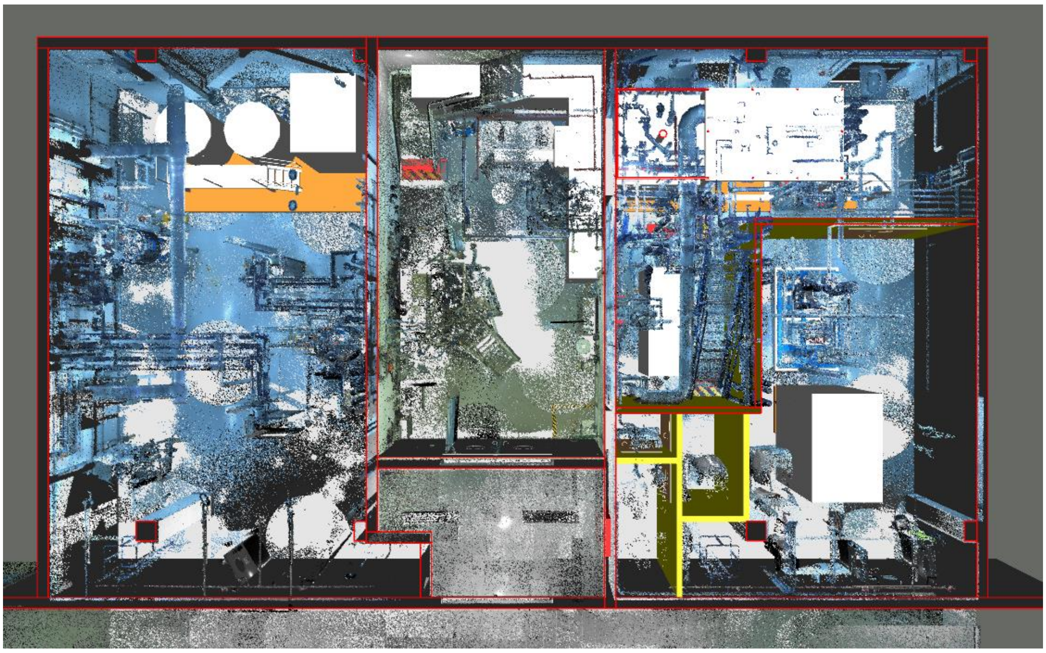

Figure 2 – Showing, from left to right, the CIP tanks, PSG, isolator, Lyophiliser, vacuum pump and new air lock change area

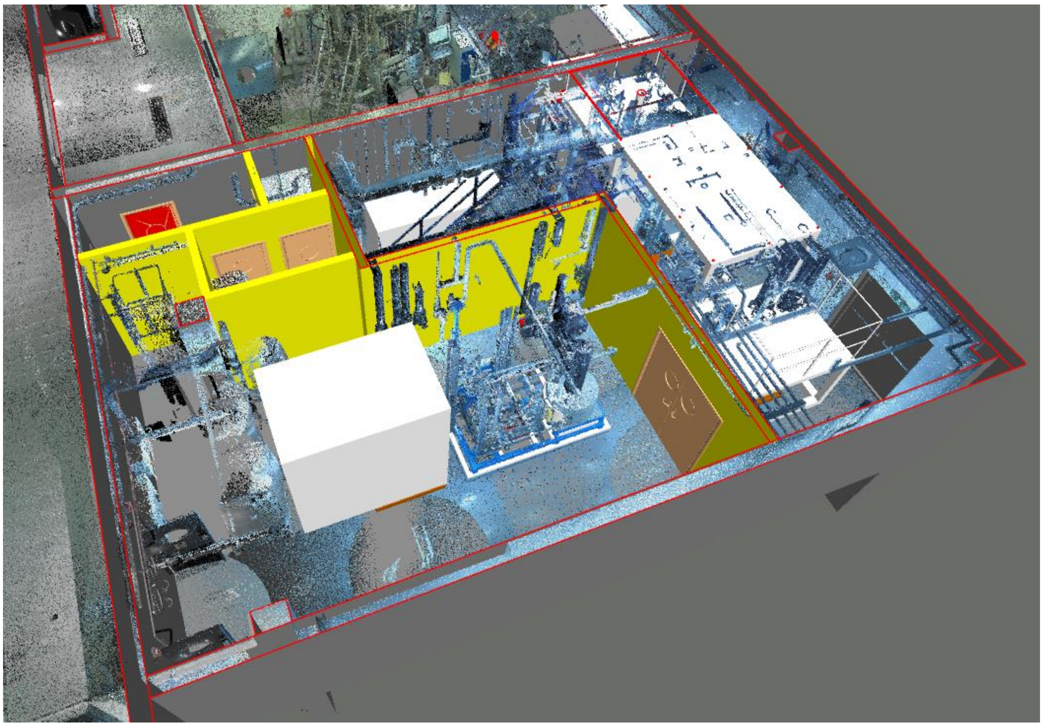

Figure 3 – Showing the subdivision of room PG.12

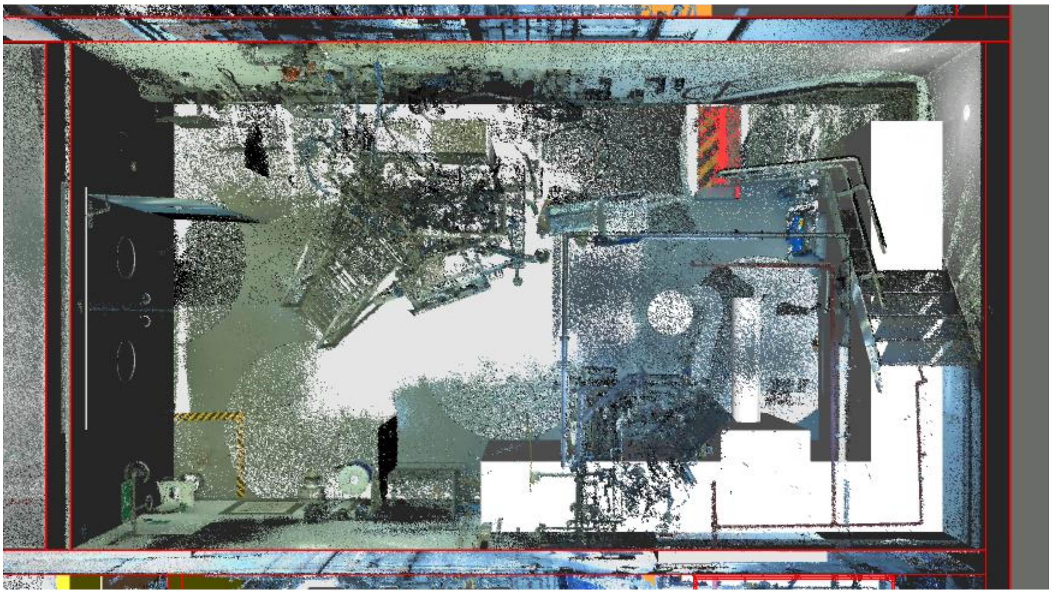

Figure 4 – Showing the isolators

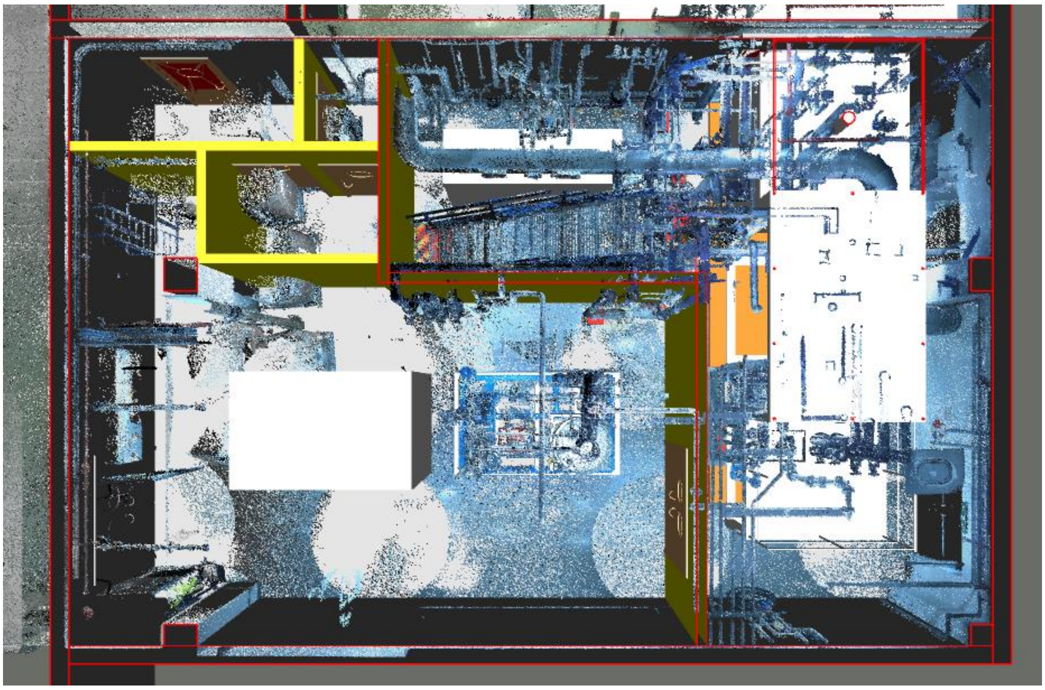

Figure 5 – Showing the Lyophiliser and relocated DVP01 vacuum pump

6.4.1 HPLC Equipment Layout (PG.05)

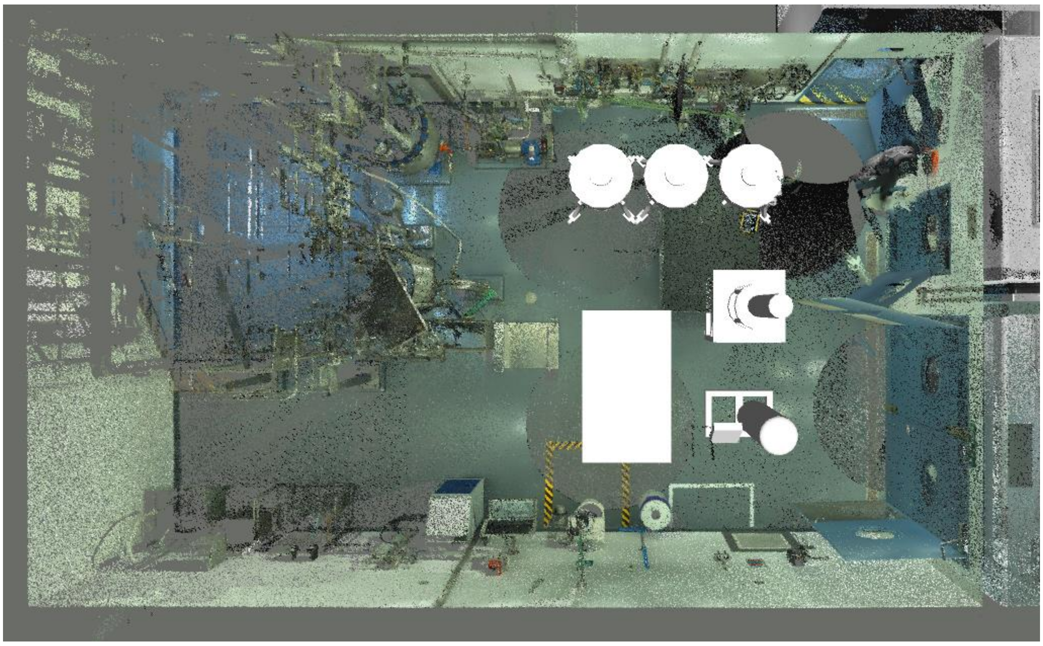

Figure 6 – Showing the layout of the HPLC including the mobile tanks (x 3), pump skid, DAC 300 and slurry tank

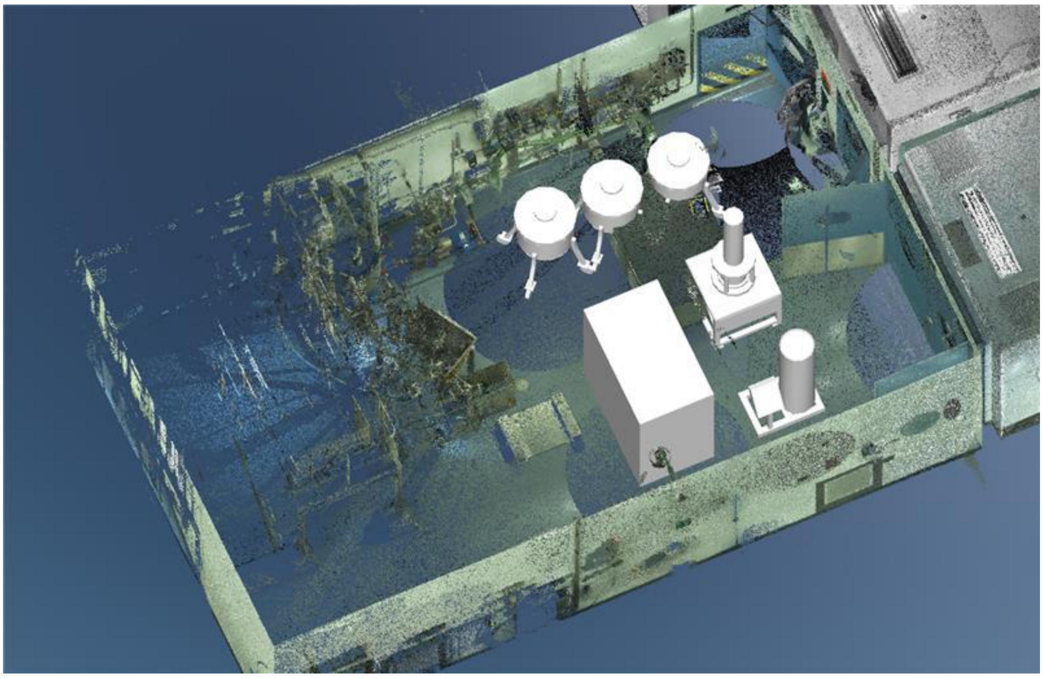

Figure 7 – Showing the 3D view of the HPLC in PG.05

7.0 HVAC

The HVAC system would be modified in accordance with the building modification namely the incorporation of a secondary change area in PG.11, and the segregation of the room in PG.12.

We would maintain the current pressure regime from the airlock PG.11 AL through the new secondary change area.

Where PG.12 is to be segregated, we would maintain the current air change rate in each area with new ductwork, grilles and volume control dampers. The grilles would match the existing.

All new supply air ductwork would be thermally insulated.

The dehumidifier discharge regeneration air would be hot and humid, so we have allowed to duct this to outside at ground floor level. The duct would be installed with a slight fall to outside to allow for any moisture to run to outside. A pipework trap would be incorporated to allow water to be piped to low level external to the building.

To ensure that the room pressure regime is not affected, we would provide a duct from outside at ground floor level. This duct would connect directly to the dehumidifier and would be thermally insulated to prevent condensation on cold days.

Before the work was commenced, the existing room air volumes would be checked to confirm that the design air volumes and air change rates are being achieved.

The existing AHUs serving the affected area would also be checked to highlight any potential issues. A report would be prepared with prices for any remedial work that was required.

On completion of the work, the system would be rebalanced to achieve the required conditions and a report issued.

8.0 CONTROL / INSTRUMENTATION

8.1 Control System

The new items of equipment will be integrated into the existing Delta V – Charms IO cabinet on the first floor. Asymchem have provided existing license allocation vs allowances to verify that there is enough capacity in the system to accommodate a small number of controls signals.

Each packaged item of equipment is anticipated to have its own self-contained control system. Details of which are not currently available. However, it is anticipated that there will be a small number of signals from the equipment items such as diagnostic fault indications and enable/disable signals etc.

8.2 Instrumentation

Instrumentation outside the equipment package scope (e.g., holding-tank level measurement) will be integrated into the existing DeltaV CHARMS I/O. All signals expected to be IS, including those from packaged equipment, will be marshalled via a local instrument junction box (IJB). Each IJB will be provided with sufficient terminals for all signals in the area inclusive of valve demand and feedback signals. The IJBs will be custom-manufactured enclosures, certified Ex e for hazardous-area use in accordance with BS 60079-14. In the project CAPEX each of the non-package tanks has been allowed a level switch a temperature transmitter and a level transmitter.

8.3 Controls Cabling

Cabling, routed via a dedicated 300mm galvanised steel ladder rack in the void space will be 20 pair 0.75mm² multicore cable Low Smoke and Fume (LSF) to each IJB where upon an allowance has been made for 80 off 10m lengths of 2 pair cables for connecting the final packages/instruments as appropriate. Cables will be marked with unique identification including an indication of the cable's origin and destination. Intrinsically safe Controls cabling will be installed on dedicated containment separately from any power circuits.

9.0 ELECTRICAL

9.1 Disinvestment

In order to optimise the layout to accommodate new equipment there are some items of equipment which have a requirement to be removed or relocated. These are detailed below.

- Pan dryer PD01
- Centrifuge C01
- Vacuum pump DVP01 (relocation)

The electrical disinvestment process will involve a systematic removal or relocation of specified equipment to facilitate the installation of new machinery. Firstly, the pan dryer PD01 and centrifuge C01 will be disconnected from their respective power supplies and carefully removed from the operational area. The vacuum pump DVP01 will be relocated to a designated position. Items being removed will have all cabling disinvested as far back as possible preferably achieving complete removal. Any cable containment that becomes empty that will not be reused by this project will be removed. For items being relocated, all electrical connections will be safely rerouted and compliant with current safety standards. Coordination with site engineers and adherence to established protocols will ensure minimal disruption to ongoing operations.

9.2 Electrical power requirements

Through discussions with Asymchem it has been established that the following items of equipment each require a power supply, MCC-P – Utilities and process supplies has available feeder sections which can be used for the purpose of feeding these items of equipment.

- Lyophiliser – 58kW
- Prep HPLC – 21 kW
- HPLC Transfer pump – TBC kW (allowed for 4.5kW)
- Lyophiliser Waste Pump – TBC kW (allowed for 4.5kW)
- HPLC Waste Pump- TBC kW (allowed for 4.5kW)
- Lyophiliser CIP Pump TBC kW (allowed for 4.5kW)

The Lyophiliser has the most significant power demand. Hence this shall occupy the space vacated by the centrifuge. However, this feed compartment would require the fuses uprated to allow for the increase. The power cable feeder to the Lyophiliser will be $35mm^2$ 4 core with a separate $16mm^2$ CPC.

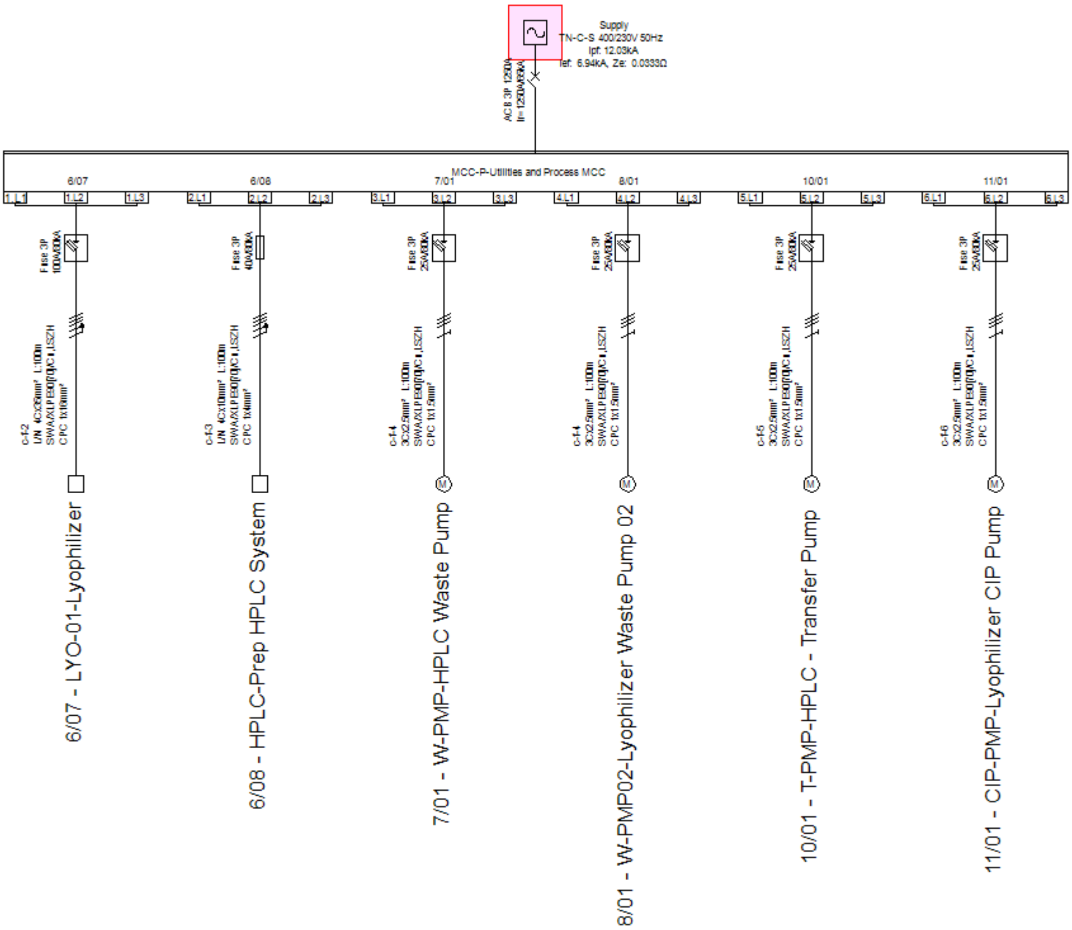

Figure 8 - Representation of the new circuits to be connected to MCC-P

9.3 Electrical containment

RB plant have not been able to ascertain whether there is any spare capacity on existing cable containment within the void space during the feasibility study. As a result we recommend that new dedicated containment is installed for the purposes of the new equipment to be installed. The costs of the containment have been allowed for in the CapEx. If in later stages of the design it is determined that there is spare space on the existing containment, then this can be used, however, grouping factors appropriate to the existing cables on the tray must be applied in line with BS 7671 18th edition amendment 4.

9.4 Lighting

The internal partition being included to form the change room and ATEX Zone limitation has resulted in a need to re-evaluate existing lighting in the former centrifuge area. Any light fittings installed will provide a lux level of 500 lux at 0.8 metres above finished floor level for the new or modified rooms. Verification that this lux level has been achieved will be carried out in later stages of design. Light fittings within ATEX areas shall be suitably rated for those areas. Typically using EXd (flameproof) and Exe (increased safety) methods of protection. Figure 1 shows the approximate lighting layout for the Lyophiliser

/CIP area. With existing fittings showing in blue, Items for removal in red and new fittings in magenta. There are no anticipated lighting changes needed in the column room.

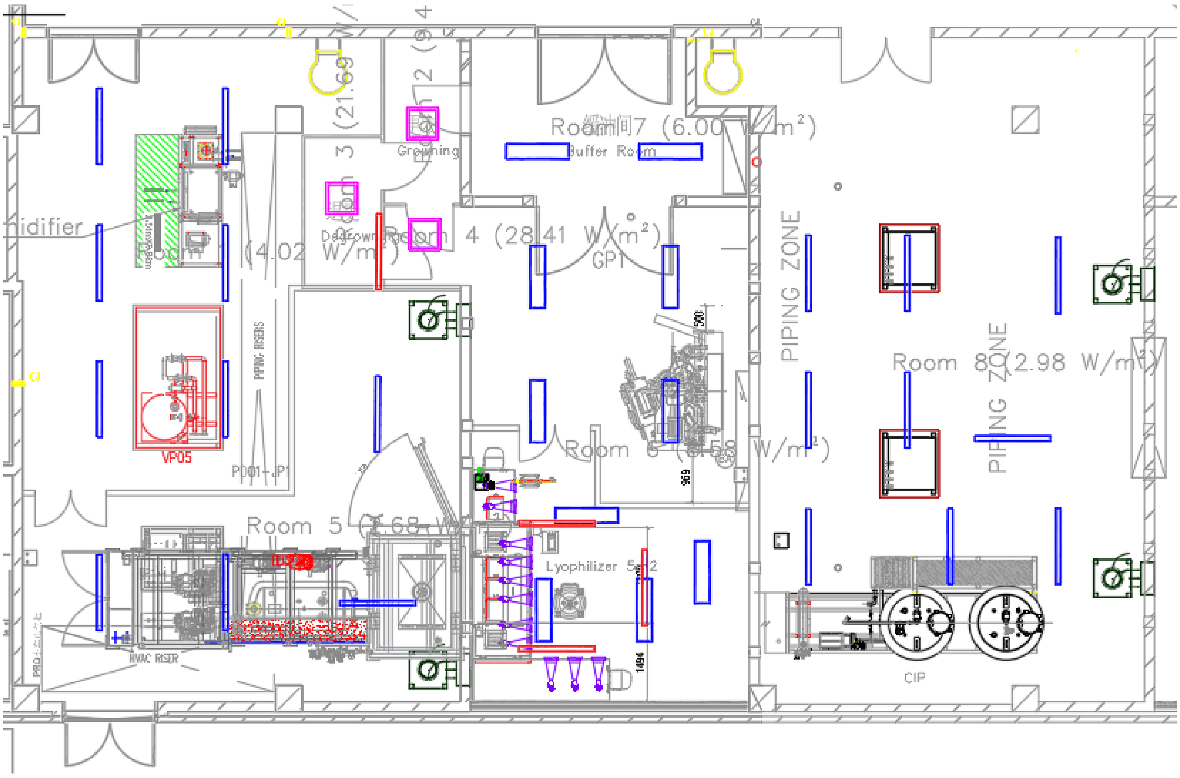

Figure 9 - Approximate Lighting Layout showing articles for removal, installation and relocation

10.0 CIVIL & STRUCTURAL WORKS

As outlined above, the installation of the Lyophiliser necessitates the removal of centrifuge C01 and pan dryer PD01 from PG.11/PG.12, along with the relocation of vacuum pump DVP01 within PG.12 (refer to the layout below).

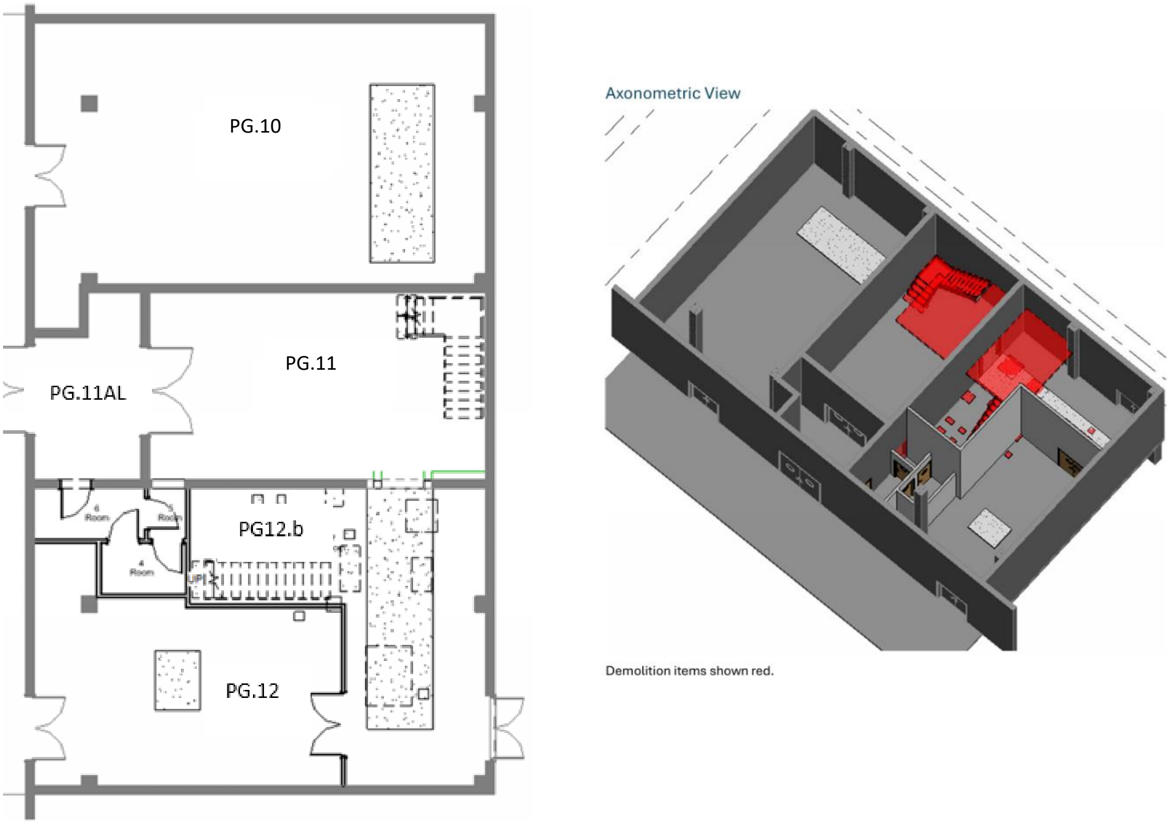

Figure 9 – Showing the new proposed room layout and the demolition elements

Within existing Room PG.12, the centrifuge, receiver vessel, and vacuum pump will be carefully and sympathetically removed and returned to Asymchem. Vacuum pump DVP01 will be relocated.

Following the removal of these equipment items, the mezzanine platform and associated access staircase will be demolished. A number of existing concrete plinths within these rooms will also be removed.

Room PG.12 will be subdivided to form an additional room, herein referred to as PG.12b. This will be constructed using a metal stud partition system incorporating double access doors. New plinths will be installed within both PG.12 and PG.12b to accommodate the associated equipment.

A new breakout opening will be created between PG.11 and PG.12b. This will include the installation of a 203UC 52 lintel beam, supported by columns, with blockwork formed above the opening. Two existing concrete-encased columns will be removed to facilitate the formation of the enlarged opening.

Within Room PG.11, centrifuge components will be removed from the mezzanine level. Following their removal, the stainless steel mezzanine platform, staircase, and associated plinth will be dismantled and removed. The safety shower and eyewash station will also be relocated as part of the works.

Gullies have been allowed for to accommodate discharge from the CIP unit in PG.10 and from Room 4. These will be connected to the existing effluent pipework using a Chemsafe Pipex system.

An airlock entry/exit system will be constructed in accordance with the arrangements shown for Rooms 4, 5, and 6, with door openings as indicated on the layout. The rooms will be formed using a metal stud partitioning system.

External to the building, in the North Road area, a new concrete bund will be constructed for the waste tank using C35 concrete, incorporating a reinforced base slab.

All structural steelwork will be developed during the FEED stage and designed in accordance with the relevant British Standards.

11.0 CONCLUSION

Based on the information available to date, this feasibility study confirms that the proposed Lyophiliser can be accommodated within Rooms PG.10, PG.11, and PG.12, with the inclusion of a new changing area airlock. An outstanding query remains regarding the overall length of the dehumidification unit; this will require further engagement with the vendor during the FEED stage to determine whether the footprint can be reduced. From a mechanical, electrical, HVAC, and civil perspective, no constraints have been identified that would prevent the development of a compliant design.

For the installation of the new HPLC unit in PG.05, the reduction in the size of the pump skid has improved access around the mobile tanks and column. No issues have been identified at the feasibility stage that will prevent progression to the subsequent design phase.

As outlined in the programme section of this report, the Lyophiliser lead time of approximately 8–10 months is the primary driver in determining the overall installation and commissioning timeline. Early placement of orders will be critical to maintaining the shortest possible programme. Further engagement with the vendor will also be important in assessing opportunities to reduce this lead time.

Multiple programme strategies are available for the HPLC. The current schedule prioritises procurement of the Lyophiliser, followed by the HPLC. Under this sequencing, a Q3 2027 operational start date for the HPLC remains achievable; however, the lyophiliser is more likely to be operational in Q4 2027.